

Five Quantum Variable Document Sets --- r_j , d_g , F_U , f_{feedback} , Ω_g , $f_{\text{Heaviside}}$, H_{SCm} ,

Daniel T. Murphy

PAPER_749: Five Quantum Variable Document Sets --- r_j , d_g , F_U , f_{feedback} , Ω_g , $f_{\text{Heaviside}}$, H_{SCm} , λ_i , M_{bh} , μ_j , γ , E_{react}

Author: Daniel T. Murphy

Framework: Universal Quantum Field Superconductive Framework (UQFF)

Session: 180 continuation | v5.38

Date: 2025

CP4 Class: #333 --- FiveQuantumVariableSetsUQFFCalculator

Abstract

Three sets of five quantum variable documents (15 variables total) were assimilated into the UQFF knowledge base during the Compression Cycle 2 thread. This paper consolidates all 15 variables with their equations, canonical values, and roles within the Unified Field Strength F_U formula. The variables span spatial distances (r_j , d_g), field strengths (F_U , μ_j), dynamics (Ω_g , f_{feedback} , γ), and operational parameters ($f_{\text{Heaviside}}$, H_{SCm} , λ_i , M_{bh} , E_{react}). Together they define the complete parameterization of the UQFF for galactic-scale applications.

1. Introduction

Multiple document sets uploaded to the Grok UQFF thread on May 08, 2025 comprised 15 individual quantum variable definition documents, each providing:

- Variable symbol and definition
- Value and units
- Role in U_m , U_{gi} , U_{bi} , F_U equations
- Example calculation for the Sun at $t=0$, $t_n=0$

This paper assimilates all 15 variables as a unified reference set.

2. Set A: Spatial and Field Variables (r_j , d_g , F_U , f_{feedback} , Ω_g)

r_j --- Magnetic String Distance
$$r_j = 1.496 \times 10^{13} \text{ m} = 100 \text{ AU}$$
Role: Distance along j -th magnetic string path (denominator in U_m and U_{g3})

$$U_m = \frac{\mu_j}{r_j} \rightarrow U_m \approx 2.28 \times 10^{65} \text{ J/m}^3$$
 U_{g3} :

$$k_3 \cos(\sigma_j B_j \cos(\omega_s \cdot t \cdot \pi)) \cdot P_{\text{core}} \cdot E_{\text{react}} \approx 1.8 \times 10^{49} \text{ J/m}^3$$

d_g --- Galactic Center Distance
$$d_g = 2.55 \times 10^{20} \text{ m} \approx 27,000 \text{ light-years}$$
Role: Distance from Sun to Milky Way center (Sgr A* reference)

$$U_{bi} = -\beta_i \cdot U_g \cdot \Omega_g \cdot (M_{bh}/d_g) \cdot (1 + \epsilon_{sw} \cdot \rho_{vac,sw}) \cdot U_{UA} \cdot \cos(\pi \cdot t_n)$$
 M_{bh}/d_g
$$= 8.15 \times 10^{36} / 2.55 \times 10^{20} \approx 3.20 \times 10^{16} \text{ kg/m}$$
 U_{b1}
$$\approx -1.94 \times 10^{27} \text{ J/m}^3$$
 U_{g4} :

$$k_4 \cdot \rho_{vac, [SCm]} \cdot (M_{bh}/d_g) \cdot e^{(-\alpha t)} \cdot \cos(\pi \cdot t_n) \cdot (1 + f_{\text{feedback}})$$
 U_{g4}
$$\approx 2.50 \times 10^{-20} \text{ J/m}^3$$

F_U --- Unified Field Strength
$$F_U = \sum_i [k_i \cdot U_{gi} - \beta_i \cdot U_g \cdot \Omega_g \cdot (M_{bh}/d_g) \cdot E_{\text{react}}] + \sum_j [\frac{\mu_j}{r_j} \cdot (1 - e^{(-\gamma t)}) \cdot \cos(\pi \cdot t_n)] \cdot \hat{\phi}_j + (g_{\mu\nu} + \epsilon_{\text{etal}} \cdot T_s^{(\mu\nu)})$$
At $t=0$, Sun: $F_U \approx U_m \approx 2.28 \times 10^{65} \text{ J/m}^3$ (U_m dominates)

f_{feedback} --- AGN Feedback Factor $f_{\text{feedback}} = 0.1$ (for $\Delta MBH = 1 \text{ dex}$ AGN feedback)

Role: Scales AGN feedback in U_{g4}

With $f_{\text{feedback}} = 0.1$: $U_{g4} \approx 2.50 \times 10^{-20} \text{ J/m}^3$

Without ($f_{\text{feedback}} = 0$): $U_{g4} \approx 2.27 \times 10^{-20} \text{ J/m}^3$

Feedback effect: $\sim 10\%$ increase $\rightarrow$ important for galaxy evolution modeling

Ω_g --- Galactic Spin Rate
$$\Omega_g = 7.3 \times 10^{-16} \text{ rad/s}$$
Role: Milky Way angular velocity (appears in U_{bi} buoyancy term)
 Rotational period: $T = 2\pi / \Omega_g \approx 8.61 \times 10^{15} \text{ s} \approx 2.73 \times 10^8 \text{ yr (galactic year)}$

3. Set B: Operational Parameters ($f_{\text{Heaviside}}$, i , H_{SCm} , λ_i , j)

$f_{\text{Heaviside}}$ --- Heaviside Component Fraction
$$f_{\text{Heaviside}} = 0.01$$
Role: Scales threshold-activated nonlinear effects in U_m
Effect in U_m : $(1 + 10^{13} \cdot f_{\text{Heaviside}}) = (1 + 10^{11})$
This amplifies U_m by factor $\sim 10^{11}$
Without $f_{\text{Heaviside}}$: $U_m \approx 2.28 \times 10^{54} \text{ J/m}^3$
With $f_{\text{Heaviside}}$: $U_m \approx 2.28 \times 10^{65} \text{ J/m}^3 \rightarrow$ canonical value

i --- Gravity Index `` $i \in \{1,2,3,4\}$ (integer index for $U_{g1}, U_{g2}, U_{g3}, U_{g4}$)

Role: Indexes the 4 universal gravity components in F_U summation

$$\Sigma(k_i \cdot U_{gi}) = k_1 \cdot U_{g1} + k_2 \cdot U_{g2} + k_3 \cdot U_{g3} + k_4 \cdot U_{g4}$$

At $t=0$, Sun: $\approx 1.42 \times 10^{53} \text{ J/m}^3$ (U_{g2} dominant)

H_{SCm} --- Heliosphere Thickness Factor $\begin{aligned} & H_{SCm} \sim 1 \text{ (dimensionless)} \parallel \& \text{ Role: Scales heliospheric thickness in } U_{g2} \parallel \& U_{g2} = k_2 \cdot (\rho_{vac}, [UA] + \rho_{vac}, [SCm]) \cdot M_s / r^2 \cdot S(r - R_b) \cdot (1 + \Delta_{sw} \cdot v_{sw}) \cdot H_{SCm} \cdot E_{react} \parallel \& \text{ With } H_{SCm} = 1.0: U_{g2} \approx 1.18 \times 10^{53} \text{ J/m}^3 \parallel \& \text{ With } H_{SCm} = 1.1: U_{g2} \approx 1.30 \times 10^{53} \text{ J/m}^3 \text{ (+10\% heliosphere thickening)} \end{aligned}$

λ_i --- Inertia Coupling Constant `` $\lambda_i = 1.0$ (uniform for all i)

Role: Scales Universal Inertia U_i in F_U

$$U_i = \lambda_i \cdot \rho_{vac}, [SCm] \cdot \rho_{vac}, [UA] \cdot \omega_s(t) \cdot \cos(\pi \cdot t_n) \cdot (1 + f_{TRZ})$$

$$= 1.0 \cdot 7.09 \times 10^{-37} \cdot 7.09 \times 10^{-36} \cdot 2.5 \times 10^{-6} \cdot 1 \cdot 1.1$$

$$\approx 1.38 \times 10^{-47} \text{ J/m}^3$$

$$\text{Net contribution: } -\lambda_i \cdot U_i \cdot E_{react} \approx -0.138 \text{ J/m}^3$$

j --- Magnetic String Index `` $j = \text{integer index for magnetic strings in } U_m \text{ and } U_{g3}$

Role: Indexes individual magnetic field strings

Σ_j acts over all contributing magnetic strings at the field point

At solar scale: single dominant string ($j=1$)

At galactic scale: multiple strings possible

4. Set C: Dynamical Variables (M_{bh} , μ_j , P_{core} , t_n , π) and (γ , E_{react} , f_{quasi} , R_b)

M_{bh} --- Black Hole Mass (Sgr A*) $\begin{aligned} & M_{bh} = 8.15 \times 10^{36} \text{ kg} \approx 4.1 \times 10^6 M_{sun} \parallel \& \text{ Role: Sgr A* mass scaling galactic gravitational field } \parallel \& \text{ Appears in } U_{bi} \text{ and } U_{g4} \text{ as } M_{bh}/d_g \text{ ratio} \end{aligned}$

μ_j --- Magnetic Moment (time-dependent) $\begin{aligned} & \mu_j(t) = (103 + 0.4 \cdot \sin(\omega_c \cdot t)) \cdot 3.38 \times 10^{20} \text{ T} \cdot \text{m}^3 \parallel \& \omega_c = 2\pi / (3.96 \times 10^8 \text{ s}) \text{ (solar magnetic cycle frequency)} \parallel \& \text{ At } t=0: \mu_j = 103 \times 3.38 \times 10^{20} = 3.38 \times 10^{23} \text{ T} \cdot \text{m}^3 \parallel \& \text{ At } t=1000 \text{ days: } (1 - e^{(-\gamma t)}) \cdot \cos(\pi \cdot t_n) \approx 0.049 \text{ to } U_m \text{ scales accordingly} \end{aligned}$

$\end{aligned}$ $\$$

$\$ \gamma$ --- Reciprocation Decay Rate $\gamma = 5 \times 10^{-5} \text{ day}^{-1}$

Role: Controls temporal decay of magnetic string effects in U_m

$1 - e^{(-\gamma t)} \rightarrow$ small for $t \ll 1/\gamma \approx 20,000$ days

At $t = 1000$ days: $1 - e^{(-0.05)} \approx 0.049$ (still growing)

E_{react} --- Reactor Efficiency Factor $E_{\text{react}} = 1046$ & Role: Universal scaling factor in all U_{gi} and U_m terms & Relates UQFF energy densities to physical observables & This constant is the primary bridge between the $\rho_{\text{vac}} [\text{SCm}]$ density scale ($\sim 10^{-37}$) and classical physics scales.
 $\end{aligned}$ $\$$

f_{quasi} --- Quasi-Longitudinal Wave Factor $f_{\text{quasi}} = 0.01$ & Role: Scales quasi-longitudinal wave contribution in U_m & $(1 + f_{\text{quasi}}) = 1.01$ (1% correction to U_m) & Models standing waves that form quasi-longitudinal components & in plasma environments, relevant to Red Dwarf Reactor dynamics.
 $\end{aligned}$ $\$$

R_b --- Radius of Outer Field Bubble $R_b = 1.496 \times 10^{13} \text{ m} = 100 \text{ AU}$ (heliospheric termination shock)

Role: Step function boundary in U_{g2}

$S(r - R_b) = 1$ for $r \geq R_b$ (heliosphere active)

$= 0$ for $r < R_b$ (interior region, different physics)

This defines the aether-superconductive boundary layer.

P_{core} --- Planetary Core Penetration Factor $P_{\text{core}} \sim 1.0$ (Sun, stars) & $P_{\text{core}} \sim 10^{-3}$ (planets, moons) & Role: Scales magnetic string core penetration in U_{g3} & $U_{g3}(\text{Sun}) \approx 1.8 \times 10^{49} \text{ J/m}^3$ & $U_{g3}(\text{planet}) \approx 1.8 \times 10^{46} \text{ J/m}^3$ (3 orders lower)
 $\end{aligned}$ $\$$

t_n --- Negative Time Factor $t_n = t - t_0$ (allows $t_n < 0$) & Role: Time reference in oscillatory terms $\cos(\pi \cdot t_n)$ & For $t = 1000$ days, $t_n = -1$: & $\cos(\pi \cdot (-1)) = -1$ (phase reversal) & U_{gi} $\rightarrow$ negative $\rightarrow$ system in negentropic regime
 $\end{aligned}$ $\$$

5. Unified Field Strength --- Complete Parameterized Equation

With all 15 variables defined:

$\$$

$\begin{aligned}$

$$\begin{aligned}
& F_U = \sum_{i=1}^4 [k_i \cdot U_{gi} - \beta_i \cdot U_{gi} \cdot \Omega_g \cdot (M_{bh}/d_g) \cdot E_{react}] \\
& + \sum_j [\mu_j(t)/r_j \cdot (1 - e^{-(\gamma t)} \cdot \cos(\pi \cdot t_n)) \cdot \hat{\phi}_j] \\
& \cdot P_{SCm} \cdot E_{react} \cdot (1 + 1013 \cdot f_{Heaviside}) \cdot (1 + f_{quasi}) \\
& + H_{SCm} \cdot (g_{\mu\nu} + \eta \cdot T_s^{\mu\nu}) \\
& - \lambda_i \cdot \sigma_i [U_i \cdot E_{react}]
\end{aligned}$$

Where all symbols are defined by the 15 quantum variable documents above.

6. Conclusion

The 15 quantum variable documents from the thread_06Jun2025.txt provide the complete parameterization of the UQFF unified field strength F_U . Each variable has been confirmed with numerical values, equations, and solar-scale calculations. Together they establish a fully specified, quantitative framework enabling computation of F_U for any astrophysical system given mass, distance, magnetic field, and temporal parameters.

Copyright - Daniel T. Murphy, daniel.murphy00@gmail.com. UQFF Framework. PAPER_749, CP4 class #333.

Session 180 continuation v5.38.

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
> (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_U B_i$ jet
> modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3),2} \cdot \frac{G M}{r_H^2}\right)$$

where:

- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford--Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left(1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right)$$

where $\Phi_{1.25\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M-- σ correction (PAPER_1048): The phonon-corrected M-- σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and
> PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$

$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	$F_{\text{U}} B_i$ buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **BH-gravity** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_\mu \phi_{\mathrm{BH}})(\partial^\mu \phi_{\mathrm{BH}}) - V(\phi_{\mathrm{BH}}) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac, [SCm]}} \cdot f_{\mathrm{[SCm]}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\mathrm{BH}}) = \frac{1}{2} m^2 \phi_{\mathrm{BH}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{BH}}^4 + \kappa \cdot \rho_{\mathrm{vac, [SCm]}} \cdot \phi_{\mathrm{BH}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{BH}}}} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \rho_{\mathrm{vac, [SCm]}} g_{\mu\nu} + F_{U_{Bi}}/r^2 = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \rightarrow \text{DPM + ACP} \rightarrow \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \rightarrow U_{\mathrm{b, seed}} \rightarrow \text{4 forces} \rightarrow F_{U_{Bi}} \rightarrow \text{sector E-L} \rightarrow \frac{S}{\delta \phi_{\mathrm{BH}}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac, [SCm]}} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac, [SCm]}} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.191\$ (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36}$ kg/m³.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 113, \quad n_{\mathrm{channel}} = 22/26$$

Since $p_{\mathrm{DVP}} = 113$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **106 M_{BH}/M_{M_sun} yr** (quasi-normal mode ringdown):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.191	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 113$	PASS Resonant
BSH layers	26 harmonic terms $j = 1\dots 26$	$\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$\$5.0 \times 10^{-4}$ day ⁻¹	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors --- Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	1.1×10^{-52} m ⁻² (UQFF vacuum term)	1.114×10^{-52} m ⁻²	Planck 2018	PASS Consistent
Proton decay rate κ	$0.0005/\text{day}$	Γ_p suppression	< $4.17 \times 10^{-35}/\text{yr}$	Super-K 2024 PASS Consistent
UQFF buoyancy signature	$F_{U_{Bi_i}}$ unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections ($F_{U_{Bi_i}}$) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF--SM bridge.

Appendix: Session 225 Cross-References (PAPER_1000--1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204--225 extensions (PAPER_1000--1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_U_Bi_i Jet Modulation
PAPER_1010	TON618 AGN F_U_Bi_i Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator --- SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy

10 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
> Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
> see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,B\}}/F_U - 1)$
where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}] * n / 26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\}} / (1 - 2^{\{1-26\}}) * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:

- $f_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q; q)_{n^2}$
- $\phi_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q^2; q^2)_n$
- $\psi_{26}(q) = \sum_{n=1}^{26} q^{n^2} / (q; q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n (1103+26390n) W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-kappa_i * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_ScM	0.99	SCm manifold completeness
rho_ScM	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_ScM	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1\CoAnQi}.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).

References

- Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
- Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
- Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem*. Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
- Weinberg, S. (1989). *The Cosmological Constant Problem*. Rev. Mod. Phys. **61**, 1 — doi:10.1103/RevModPhys.61.1
- Fabian, A.C. (2012). *Observational Evidence of Active Galactic Nuclei Feedback*. ARA&A **50**, 455 — arXiv:1204.4114 — doi:10.1146/annurev-astro-081811-125521
- McNamara, B.R. & Nulsen, P.E.J. (2007). *Heating Hot Atmospheres with Active Galactic Nuclei*. ARA&A

45, 117 — arXiv:0709.4098 — doi:10.1146/annurev.astro.45.051806.110625

7. Heckman, T.M. & Best, P.N. (2014). *The Coevolution of Galaxies and Supermassive Black Holes*.

ARA&A **52**, 589 — arXiv:1403.4620 — doi:10.1146/annurev-astro-081913-035722